- **Trigonometria**

α	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\alpha$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\alpha$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\alpha$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	–

Archi associati

$$\sin(-\alpha) = -\sin\alpha, \qquad \cos(-\alpha) = \cos\alpha$$
$$\sin(\pi/2-\alpha) = \cos\alpha, \qquad \cos(\pi/2-\alpha) = \sin\alpha$$
$$\sin(\pi/2+\alpha) = \cos\alpha, \qquad \cos(\pi/2+\alpha) = -\sin\alpha$$
$$\sin(\pi-\alpha) = \sin\alpha, \qquad \cos(\pi-\alpha) = -\cos\alpha$$
$$\sin(\pi+\alpha) = -\sin\alpha, \qquad \cos(\pi+\alpha) = -\cos\alpha$$

Formule di addizione e sottrazione

$$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$$
$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$$
$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\beta\sin\alpha$$
$$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\beta\sin\alpha$$

Formule di duplicazione

$$\sin(2\alpha) = 2\sin\alpha\cos\alpha$$
$$\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha = 1 - 2\sin^2\alpha = 2\cos^2\alpha - 1$$

Formule di bisezione

$$\sin^2\left(\frac{\alpha}{2}\right) = \frac{1-\cos\alpha}{2}, \qquad \cos^2\left(\frac{\alpha}{2}\right) = \frac{1+\cos\alpha}{2}$$

Formule parametriche

$$\sin\alpha = \frac{2\tan(\alpha/2)}{1+\tan^2(\alpha/2)}, \qquad \cos\alpha = \frac{1-\tan^2(\alpha/2)}{1+\tan^2(\alpha/2)}$$

Formule di prostaferesi

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\sin\alpha - \sin\beta = 2\sin\left(\frac{\alpha-\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)$$
$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$
$$\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

- **Limiti notevoli**

$$\lim_n \frac{\log_a n}{n^\alpha} = 0 \quad (a>0, a\neq 1,\ \alpha>0), \quad \lim_n \frac{n^\alpha}{a^n} = 0 \quad (a>1,\ \alpha>0), \quad \lim_n \frac{a^n}{n!} = 0 \quad (a>1,\ \alpha>0), \quad \lim_n \frac{n!}{n^n} = 0$$

$$\lim_n \left(1+\frac{1}{n}\right)^n = e, \quad \lim_n \frac{\sqrt[n]{n!}}{n} = \frac{1}{e}$$

$$\lim_{x\to+\infty} \frac{\log_a x}{x^\alpha} = 0 \quad (a>0, a\neq 1,\ \alpha>0), \quad \lim_{x\to+\infty} \frac{x^\alpha}{a^x} = 0 \quad (a>1,\ \alpha>0)$$

$$\lim_{x\to0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to0} \frac{\tan x}{x} = 1, \qquad \lim_{x\to0} \frac{\arcsin x}{x} = 1, \qquad \lim_{x\to0} \frac{\arctan x}{x} = 1, \qquad \lim_{x\to0} \frac{1-\cos x}{x^2} = \frac{1}{2},$$

$$\lim_{x\to0} \frac{a^x-1}{x} = \log a \quad (a>0, a\neq 1), \qquad \lim_{x\to0} \frac{\log_a(1+x)}{x} = \frac{1}{\log a} \quad (a>0, a\neq 1)$$

$$\lim_{x\to0} \frac{(1+x)^\alpha - 1}{x} = \alpha \quad (\alpha\in\mathbb{R}), \qquad \lim_{x\to\pm\infty} \left(1+\frac{1}{x}\right)^x = e$$

- **Derivate delle funzioni elementari** ($a > 0, a \neq 1$, $\alpha \in \mathbb{R}$)

$D(k) = 0, k \in \mathbb{R}$; $\quad D(x^\alpha) = \alpha x^{\alpha-1}$; $\quad D(a^x) = a^x \log a$; $\quad D(\log_a |x|) = \dfrac{1}{x \log a}$; $\quad D(\sin x) = \cos x$; $\quad D(\cos x) = -\sin x$

$D(\tan x) = 1 + \tan^2 x = \dfrac{1}{\cos^2 x}$; $\quad D(\arcsin x) = \dfrac{1}{\sqrt{1-x^2}}$; $\quad D(\arccos x) = -\dfrac{1}{\sqrt{1-x^2}}$; $\quad D(\arctan x) = \dfrac{1}{1+x^2}$

- **Formule di Taylor**

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + ... + \frac{x^n}{n!} + o(x^n)$$
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + ... + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$
$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4!} + ... + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$$
$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + o(x^8)$$
$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + ... + (-1)^{n+1} \frac{x^n}{n} + o(x^n)$$
$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + ... + \binom{\alpha}{n} x^n + o(x^n), \qquad \alpha \in \mathbb{R},$$
$$\text{dove } \binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}$$
$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + ... + (-1)^{n+1} \frac{(2n-3)!!}{(2n)!!} x^n + o(x^n)$$
$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + ... + (-1)^n x^n + o(x^n)$$
$$\frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{3}{8}x^2 + ... + (-1)^n \frac{(2n-1)!!}{(2n)!!} x^n + o(x^n)$$
$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} + ... + (-1)^n \frac{x^{2n+1}}{2n+1} + o(x^{2n+2})$$
$$\arcsin x = x + \frac{x^3}{6} + \frac{3}{40}x^5 + ... + \frac{(2n-1)!!}{(2n)!!} \frac{x^{2n+1}}{2n+1} + o(x^{2n+2})$$

- **Integrali immediati** ($c \in \mathbb{R}$)

$$\int x^\alpha \, dx = \frac{x^{\alpha+1}}{\alpha+1} + c \quad (\alpha \neq -1), \qquad \int \frac{1}{x} \, dx = \log|x| + c$$
$$\int a^x \, dx = \frac{a^x}{\log a} + c \quad (a > 0, a \neq 1), \qquad \int \sin x \, dx = -\cos x + c$$
$$\int \cos x \, dx = \sin x + c, \qquad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + c$$
$$\int \frac{1}{1+x^2} \, dx = \arctan x + c,$$
$$\int \frac{1}{\sqrt{x^2+1}} \, dx = \log(x + \sqrt{x^2+1}) + c = \text{settsinh}\, x + c, \qquad \int \frac{1}{\sqrt{x^2-1}} \, dx = \log(x + \sqrt{x^2-1}) + c = \text{settcosh}\, x + c,$$